

simulation allows training of future doctors in surgical procedures without the need of an actual human body. Improvements in the technology could also lead to more robust systems for performing remote surgeries.

Haptics is also used in electronic prosthetics to give patients a better sense of touch, pressure and position to imitate the complexity of a human grip.

A tight-fitting shirt with haptic actuators linked to sensors would provide pulses in different forms like pixels on a display, giving sensory impression of the surroundings or of any event to which they are linked. It would function like a whole new sense for a human, just like how we are able to currently sense light and sound in our surroundings. Interfacing this with motion-detecting cameras would allow a person to identify his surroundings without actually seeing them.

A more ambitious implementation would be the merger of new concepts of wearable consumer electronics, such as Google Glass, with haptic technology, thus allowing people to interact 'physically' with virtual objects as if they were real-world objects.

Haptics could also be extremely useful for visually or hearing impaired people. These people have superb touch-sense ability and haptics-based devices could be a boon for them.

"Imagine a computer with Braille keypad available today getting replaced with a computer with a normal keypad that is not only usable by the non-impaired people but also has haptics for each key, thus providing usability to the visually impaired users. Such advances would prove to be useful for communication and commercial devices, benefitting the physically impaired in addition to typical regular users. I don't see any application or device in WIP phase which could cater to this idea or needs but it would be interesting to see this in the near future," suggests Tamboli.

"As of now, in the gaming sector, haptic technology seems to be restrict-

ed to tactile feedback in consoles and in providing risk-free environments for medical research. However, a truly open source haptic hardware with an IDE could totally change the scenario in the same way as Arduino, Raspberry Pi and Lego NXT have changed microcontroller programming and robotics. It could totally revolutionise each and every aspect of our life in the same way as computer graphics did two decades ago. Haptics may take time to enter our daily life, but it definitely has a place in the entertainment industry and in research in life sciences and robotics," adds George Thomas, Foss evangelist and Arduino hobbyist.

The other concurrent research is being done by Anupam Varkey, head, computer department, Thapar Polytechnic, Thapar University, Patiala. It includes developing predictive kinematic positioning algorithms for haptic device control.

"There is always a delay in sensing and its inherent haptic control, however small. This is because it takes time for an actuator to work at the desired force. Once these predictive algorithms are developed, it will be possible to feel the actions much more realistically. Another benefit of this research is in developing visualisation software for use of haptics in simulation. Use of haptics in 'what-if' scenarios becomes more realistic and less computation power has to be used. If developed fully, it will be very easy to use haptics-based devices in numerous applications, especially teaching technology. An engineering student can practice maybe taking apart and putting together a whole turbine in a power generation house."

There is a limit to the range of senses that we can employ to interact with the virtual world. Audio-visual interaction has peaked and there is very little that can be added to the three-dimensional visuals and multi-channel surround sound. ●

The author is a tech correspondent at EY Bengaluru

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